



SIMATS Online Programming Lab

JAVA PROGRAMMING



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CO2-AT1-Assignment1

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QUESTIONS

Q1

Smart Document Editor using String, StringBuilder and StringBuffer

10 marks

**Q2**

Hospital Token Management System using List Interface

10 marks

**Q3**

Airport Boarding Queue Management using Queue Interface

10 marks

**Q4**

Digital Certificate Verification using Set Interface

10 marks

**Q5**

Smart Inventory Management using Map Interface

10 marks



5/5 answered

Q3

Airport Boarding Queue Management using Queue Interface

10 marks

Airport Boarding Queue Management using Queue Interface

Add passengers to the boarding queue.

Display the current queue.

Board passengers in FIFO order.

Display the next passenger without removing them.

Handle VIP passengers using a priority-based queue.

Display remaining passengers after each boarding operation.

Requirement: Use appropriate Queue implementations.

Your Code

```
1 import java.util.LinkedList;
2 import java.util.PriorityQueue;
3 import java.util.Queue;
4 import java.util.Comparator;
5
6 public class Passenger {
7     String name;
8     int priority;
```

```

9      public Passenger(String name, int priori
10         this.name = name;
11         this.priority = priority;
12     }
13     @Override
14     public String toString() {
15         return name;
16     }
17     public static void main(String[] args) {
18         Queue<String> regularQueue = new Lin
19
20         System.out.println("--- Regular Pass
21         regularQueue.add("Alice");
22         regularQueue.add("Bob");
23         regularQueue.add("Charlie");
24
25         System.out.println("Current Queue: "
26
27         System.out.println("Next to board (p
28
29         System.out.println("Boarding: " + re
30         System.out.println("Remaining Queue:
31
32         System.out.println("Boarding: " + re
33         System.out.println("Remaining Queue:
34
35         System.out.println("\n--- VIP Passen
36
37         PriorityQueue<Passenger> vipQueue =
38             Comparator.comparingInt(p -> p.p
39         );
40         vipQueue.add(new Passenger("David (E
41         vipQueue.add(new Passenger("Eva (VIP
42         vipQueue.add(new Passenger("Frank (B
43
44         System.out.println("Boarding VIP/Pri
45         while (!vipQueue.isEmpty()) {
46             System.out.println("Boarded: " +
47             System.out.println("Remaining VI
48         }
49     }
50 }

```



Input

stdin input

Output

```
--- Regular Passenger Queue (FIFO) ---  
Current Queue: [Alice, Bob, Charlie]  
Next to board (peek): Alice  
Boarding: Alice  
Remaining Queue: [Bob, Charlie]  
Boarding: Bob
```



Run



Save

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